

OLD STARS

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OLD STARS INCLUDE low-mass main-sequence stars that came into existence billions of years ago and also some high-mass stars that will explode as supernovae after existing for less than a million years. Some of the

most beautiful sights in the Milky Way are old stars undergoing their death throes.



EVOLVED STARS

It is easy to pick out the evolved red giant stars in this image of the ancient star cluster NGC 2266.

RED GIANTS

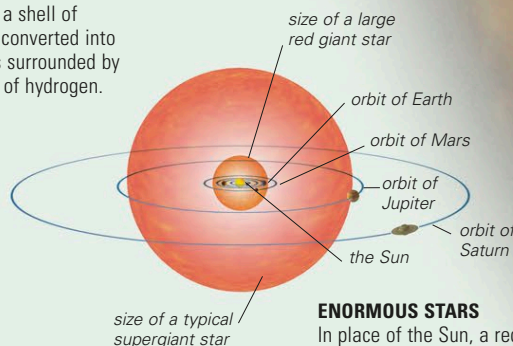
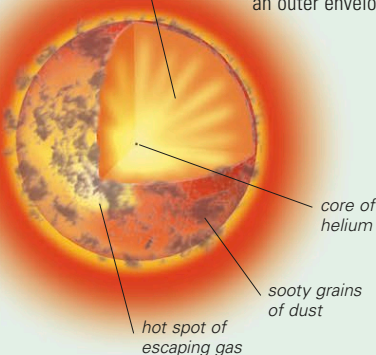
When a star has depleted the hydrogen in its core, it will start to burn the hydrogen in a shell surrounding the core. This shell gradually moves outward through the atmosphere of the star as fuel is used up. The expanding source of radiation heats the outer atmosphere, which expands and then cools. The result is a large star with a relatively low surface temperature. It remains luminous because of its huge size, though some red giants are hidden from view by extensive dust clouds.

Red giants have surface temperatures of 3,600–7,200°F (2,000–4,000°C) and radii 10–100 times that of the Sun. Because they are so large, gravity does not have much effect on their outer layers and they can lose a great deal of mass to the interstellar medium, either by stellar winds or in the form of planetary nebulae. Red giants are often variable stars; their outer layers pulsate, causing changes in luminosity (see p.282).

INSIDE A RED GIANT

A red giant's helium core is contained by an inert helium shell. Outside this zone, a shell of hydrogen is being converted into helium, and this is surrounded by an outer envelope of hydrogen.

convection cells carry heat from core to surface



ENORMOUS STARS

In place of the Sun, a red giant would reach beyond the orbit of the Earth, while a supergiant would have a radius reaching out to Jupiter's orbit.

SUPERGIANTS

Stars of very high mass expand to become even larger than red giants. Red supergiants can have radii several hundred times that of the Sun. Just like red giants, they undergo hydrogen-shell burning (see p.236) and leave the main sequence (see p.232). When they have finished hydrogen-shell burning, they collapse and the helium core reaches a high enough temperature for the helium to be converted into carbon and oxygen. Helium-core burning is briefer than hydrogen burning, and when the helium core is depleted, helium-shell burning begins. If massive enough,

further nuclear burning will occur, producing elements with an atomic mass up to that of iron. Near the end of the supergiant phase, a high-mass star will develop several layers of increasingly heavy elements. Eventually, supergiants die as supernovae.

GARNET STAR

One of the largest stars visible in the night sky, Mu Cephei or the Garnet Star is a red supergiant with a radius greater than that of Jupiter's orbit.

